

Comparative Fault Localization Analysis Report

Describer-Decider v33.0 — Full Dataset Validation (7 Own Traces + 5 Group 411 Cases)

1. Executive Summary

Metric	Our Traces (7 apps)	Group 411 Traces (5 cases)	Combined (12 apps)
Total apps evaluated	7	5	12
Total steps in dataset	90	46	136
Steps skipped (non-VLM / wait)	5	0	5
Steps analyzed	85	46	131
Ground truth failures	12	13	25
System-predicted failures	12	12	24
True Positives (TP)	12	11	23
False Positives (FP)	0	1	1
False Negatives (FN)	0	2	2
Precision	100.0%	91.7%	95.8%
Recall	100.0%	84.6%	92.0%
F1 Score	100.0%	88.0%	93.9%

Note on step counts: Our "analyzed" count counts only VLM-processed steps (skipped steps removed). For the 90 total steps in our 7 traces, 5 are skipped (Bitaksi: 1, Flight App: 3, Yemeksepeti: 1), leaving **85 analyzed**. Group 411 traces have no skipped steps: all 46 are analyzed.

Bottom line (Our Traces): v33.0 achieves **perfect precision and recall** on all 7 proprietary test cases — every real failure was detected, and no pass was incorrectly flagged. This represents a significant improvement over the previous evaluation (v2 report), which had 6 false positives from log misalignment and Rule 2 over-application. All systematic issues identified in the prior report were resolved in v33.0.

Bottom line (Group 411 Traces): Applied to an independent external dataset, the system achieves **91.7% precision and 84.6% recall (F1 = 88.0%)**. Two false negatives arise from: (1) a subtle post-transfer balance inconsistency requiring cross-step state tracking (Bank App Step 6), and (2) a missed visual mismatch at Rent-a-Car Step 8, where the discount was claimed applied by the UI but the total price ₺17,000 remained unchanged — a visible inconsistency the observer failed to flag. One false positive arises from a VLM observation error on the Passo App (button state misread as grayed-out when ticket categories were actually shown).

Combined overall F1: 93.9% — demonstrating strong generalization from the proprietary training-style traces to independently designed external test cases.

2. Ground Truth — Confirmed Real Failures

2A. Our Own Traces

App	Step	Failure Description	Evidence Type
Migros	Step 7	POST /api/v1/cart/confirm → 400 Bad Request; cart state inconsistency (local: 1 item, server: EMPTY); UI showed empty cart	Log: status: FAILED + network ERROR
Bitaksi	—	No failures	All StepRunner markers: PASSED
Clock	—	No failures	All steps progressed normally; no log errors
Flight App	Step 18	Click on 12.15→13.35 flight item did not select it; wrong flight context (21:15) loaded	Visual: screen shows 21:15 departure in POST
Flight App	Steps 20–24	Cascading content_mismatch: each verification step confirms 21:15 instead of 12:15	Visual: expected flight time never appears
QNB	Step 4	Click on Günüluk option produced no UI state change; button remained unselected	Visual: no change between PREV/POST
Yemeksepeti	Step 7	Click on Mixed (Medium) Pizza opened Mediterranean (Central) detail page	Visual + Log: wrong item in POST
Yemeksepeti	Step 14	AbstractVerification confirmed wrong pizza persists to checkout	Visual: Mediterranean (Central) in order summary
O Bilet	Step 7	POST /api/v3/bus/reserve-seat → 504 Gateway Timeout; reservation service unresponsive	Log: NET_FAIL + SYS_ERR
O Bilet	Step 8	Navigation to PassengerDetailsActivity blocked by failed reservation dependency	Log: SYS_ERR + UI_STALL

Total: 12 real failures across 7 apps.

2B. Group 411 Traces

App	Step	Failure Description	Evidence Type
Passo App	Step 10	Click on occupied (black) seat — no state change; seat remained unselectable	Visual: no change between PREV/POST

App	Step	Failure Description	Evidence Type
Bank App	Step 4	POST /transfer → 500 Internal Server Error; transfer silently failed on backend	Log: NET_FAIL + SYS_ERR
Bank App	Step 6	Dashboard shows balance of 5250.00 TL unchanged after a "successful" transfer — systematic inconsistency between displayed success and actual server rejection	Cross-step state: balance should update after Step 5 success screen
Bank App	Step 7	Abstract verification element Bakiyeyi doğrula not visible in either PREV or POST	Visual: element absent
Ticket App	Step 2	Click on Theater category — no UI response or navigation occurred	Visual: no change between PREV/POST
Ticket App	Step 3	RenderFlex overflow by 869 pixels on right side of Sports Events screen	Log: rendering exception in console
Ticket App	Step 6	Click on occupied seat A3 — system error Seat unavailable ; seat not selectable	Log: SYS_ERR; Visual: no change
Ticket App	Step 10	POST /api/v1/payment → 500 Internal Server Error; payment could not be processed	Log: NET_FAIL + SYS_ERR
Rent-a-Car	Step 5	Click on Kirala for Mercedes-Benz G-Class blocked — corporate account required for intercity Izmir drop-off	Log: SYS_ERR
Rent-a-Car	Step 8	Promo code DISCOUNT20 entered and UI claimed discount applied, but total price remained ₺17,000 — visible price-discount inconsistency missed by the observer	Visual: discount message shown but total unchanged
Rent-a-Car	Step 9	Discount message " Tebrikler! %20 indirim uygulandı! " appeared but total remained ₺17,000 — content mismatch on amount	Visual: amount unchanged
Rent-a-Car	Step 10	POST /payment → 504 Gateway Timeout on Izmir regional gateway	Log: NET_FAIL + SYS_ERR
Wikipedia	Step 4	Instruction target Osmanlı padişahı 3. Murad tahta çıktı seç. not present in any visible UI element	Visual: element absent; instruction unrelated to current page context

Total: 13 real failures across 5 cases.

3. Part A — Our Own Traces: Case-by-Case Breakdown

App 1 — Migros · 8 steps · Log: nested (preassigned)

Scenario: App splash → main dashboard → Categories tab → search "Raffaello" → add to cart → select delivery address → view cart → confirm order.

Step	Action / Element	System Result	Ground Truth	Assessment
0	waitUntil — Migros Splash Logo	PASS	Pass ✓	TN
1	waitUntil — Service Selection Dashboard	PASS	Pass ✓	TN
2	click — Categories tab	PASS	Pass ✓	TN
3	type — Search "Raffaello"	PASS	Pass ✓	TN
4	click — Add to Cart (Ferrero Raffaello 150G)	PASS	Pass ✓	TN
5	click — Yurt delivery address	PASS	Pass ✓	TN
6	click — Cart icon	PASS	Pass ✓	TN
7	click — Confirm button	FAIL (system_error)	Fail X	TP ✓

Result: 1 failure detected, 1 real failure → Perfect precision and recall.

True Positive — Step 7: The cart confirmation failure was definitively detected. Log evidence: `POST /api/v1/cart/confirm → 400 Bad Request (Cart is empty), STATE_MISMATCH: Local cart state shows 1 item but session validation returned EMPTY`. The visual observer confirmed the cart appeared empty in POST with the Confirm button absent. The `[NET_FAIL, SYS_ERR]` flags correctly elevated the verdict to `system_error`. Classified as **ISOLATED FAILURE** — no upstream cause was found, correctly reflecting that the root is a backend session state inconsistency, not a cascade from prior steps.

Improvement vs. previous evaluation (v2 report): The prior report showed 3 predicted failures (Steps 0, 3, 7) with 2 false positives. v33.0 eliminated both FPs: (1) Step 0's `waitUntil` + presence check no longer triggers Rule 2 strict value verification, and (2) Step 3's search result OCR artifact ("Rafeollo") is no longer misclassified as a content mismatch. The invalid chain `Step 0 → Step 3 → Step 7` is gone; the real failure at Step 7 now stands correctly as an isolated backend event.

App 2 — Bitaksi · 9 steps (1 skipped) · Log: flat_assertions

Scenario: App launch (waitUntil, skip) → tap "Where to?" → save home address → select Lux ride → profile info → save profile → trip history → payment methods → delete card.

Step	Action / Element	System Result	Ground Truth	Assessment
0	waitUntil — Main map screen [VLM=false]	SKIPPED	Pass (skip)	Correct skip
1	click — "Where to?" search input	PASS	Pass ✓	TN
2	click — Save (home address)	PASS	Pass ✓	TN
3	click — Lux ride option	PASS	Pass ✓	TN
4	click — Profile information	PASS	Pass ✓	TN
5	click — Save button	PASS	Pass ✓	TN

Step	Action / Element	System Result	Ground Truth	Assessment
6	click — Trip history	PASS	Pass ✓	TN
7	click — Payment methods	PASS	Pass ✓	TN
8	click — Delete card	PASS	Pass ✓	TN

Result: 0 failures detected, 0 real failures → Perfect TN.

App 3 — Clock · 7 steps · Log: flat_indexed

Scenario: Enable alarm toggle → navigate to Stopwatch tab → start stopwatch → reset stopwatch → scroll minute picker → pause timer → navigate to World Clock tab.

Step	Action / Element	System Result	Ground Truth	Assessment
0	click — Alarm toggle (08:17)	PASS	Pass ✓	TN
1	click — Stopwatch tab	PASS	Pass ✓	TN
2	click — Start button	PASS	Pass ✓	TN
3	click — Reset button	PASS	Pass ✓	TN
4	scroll — Minute picker	PASS	Pass ✓	TN
5	click — Pause button	PASS	Pass ✓	TN
6	click — World Clock tab	PASS	Pass ✓	TN

Result: 0 failures detected, 0 real failures → Perfect TN.

App 4 — Flight App · 25 steps (3 skipped) · Log: dynamic (keyword-based)

Scenario: App splash → waitUntil (×2, skip) → dismiss X icon → verify login screen → guest session → allow notification permission → select departure city Nereden → pick İzmir → select destination → pick Ankara → open departure calendar → select date 21 → search flights → wait (skip) → select flight class → pick flight time → search → click flight 12.15→13.35 → select BASIC package → abstractVerification (×5 steps).

Step	Action / Element	System Result	Ground Truth	Assessment
0	waitUntil — "Tüm seyahatin tek uygulamada"	PASS	Pass ✓	TN
1–2	waitUntil steps [VLM=false]	SKIPPED	Pass (skip)	Correct skip
3	click — Çarpı (X icon)	PASS	Pass ✓	TN
4	verify — Login screen	PASS	Pass ✓	TN
5	click — Üye olmadan devam et	PASS	Pass ✓	TN

Step	Action / Element	System Result	Ground Truth	Assessment
6	click — İzin Ver (notifications)	PASS	Pass ✓	TN
7	click — Nereden (departure city)	PASS	Pass ✓	TN
8	click — İzmir	PASS	Pass ✓	TN
9	click — Arrival city input	PASS	Pass ✓	TN
10	click — Ankara	PASS	Pass ✓	TN
11	click — Departure date field	PASS	Pass ✓	TN
12	click — Calendar date 21	PASS	Pass ✓	TN
13	click — Search button (UCUZ BİLET BUL)	PASS	Pass ✓	TN
14	wait [action=wait]	SKIPPED	Pass (skip)	Correct skip
15	click — Sort by departure time	PASS	Pass ✓	TN
16	click — Evening flights filter	PASS	Pass ✓	TN
17	click — Show more flights	PASS	Pass ✓	TN
18	click — 12.15→13.35 flight item	FAIL (content_mismatch)	Fail ✗	TP ✓
19	click — BASIC package option	PASS	Pass ✓	TN
20	abstractVerify — flight time 12:15–13:35	FAIL (content_mismatch)	Fail ✗	TP ✓
21	abstractVerify — BASIC + 12:15	FAIL (content_mismatch)	Fail ✗	TP ✓
22	abstractVerify — flight time 12:15–13:35	FAIL (content_mismatch, caused by Step 21)	Fail ✗	TP ✓
23	abstractVerify — 12:15 departure + BASIC	FAIL (content_mismatch, caused by Step 22)	Fail ✗	TP ✓
24	abstractVerify — 12:15–13:35 final confirm	FAIL (content_mismatch, caused by Step 23)	Fail ✗	TP ✓

Result: 6 failures detected, 6 real failures → Perfect precision and recall.

True Positive — Step 18 (Root Failure): The click on the 12.15→13.35 flight item loaded a different flight context. The visual observer noted the flight class selection overlay appeared with no sign of the targeted time slot. Classified as **content_mismatch** — the element was present in PREV (list view) but the POST screen showed the wrong flight context (21:15 departure).

True Positives — Steps 20–24 (Cascading Chain): All five downstream verification steps correctly flagged the same root cause: the screen consistently shows a 21:15–22:35 departure/arrival pair instead of the expected 12:15–13:35. The causal chain `Step 18 → Step 20 → Step 21 → Step 22 → Step 23 → Step 24` was correctly constructed by `find_likely_cause()` using element overlap and temporal proximity scoring across all six steps.

Notes on dynamic log slicing: The flight app uses raw/dynamic log mode with keyword-based log assignment (5 entries per step consistently). The log evidence was supportive but not critical — the failure was clearly visible in the screenshots. The dynamic slicer's performance here validates that keyword matching (keyword mode) produces relevant evidence even without pre-assigned step buckets.

App 5 — QNB · 13 steps · Log: flat_indexed

Scenario: Open Kartlar → navigate Kartlar → tap Banka Kartlarım → tap Back → open Limit Geçerlilik Süresi → select Günlük → close dropdown → navigate back → open Banka Kartı Ayarları → tap Güncelle → enter limit value → tap Uygula → confirm (Onayla).

Step	Action / Element	System Result	Ground Truth	Assessment
0	click — Kartlar menu	PASS	Pass ✓	TN
1	click — Banka Kartlarım	PASS	Pass ✓	TN
2	click — Back (to main Kartlar)	PASS	Pass ✓	TN
3	click — Limit Geçerlilik Süresi	PASS	Pass ✓	TN
4	click — Günlük option	FAIL (action_failed)	Fail ✗	TP ✓
5	click — Günlük (select)	PASS	Pass ✓	TN
6	dropdown closes; Günlük confirmed	PASS	Pass ✓	TN
7	click — Back (to Kartlar menu)	PASS	Pass ✓	TN
8	click — Banka Kartı Ayarları	PASS	Pass ✓	TN
9	click — Güncelle button	PASS	Pass ✓	TN
10	tap — İşlem Limitiniz input field	PASS	Pass ✓	TN
11	tap — Uygula button	PASS	Pass ✓	TN
12	tap — Onayla (confirmation popup)	PASS	Pass ✓	TN

Result: 1 failure detected, 1 real failure → Perfect precision and recall.

True Positive — Step 4: The click on the `Günlük` option produced no observable UI state change — the option remained visible but neither selected nor highlighted. Classified as `action_failed`. The log evidence (entry: "tapped Limit Geçerlilik Süresi modal opened") confirmed a modal was triggered but the `Günlük` item itself was not responsive. This is correctly isolated (not cascaded) since Step 5 successfully selected `Günlük` using a different interaction.

App 6 — Yemeksepeti · 15 steps (1 skipped) · Log: nested (no logs field)

Scenario: Splash screen → waitUntil sign-up (skip) → close Flash Deals banner → open address bar → select delivery address → search "little ceasers pizza" → open Little Caesars → select Mixed (Medium) Pizza → select Normal Dough → select Less Pizza Sauce → remove ingredients → add to cart → view cart → confirm basket → abstractVerification of checkout.

Step	Action / Element	System Result	Ground Truth	Assessment
0	waitUntil — Yemeksepeti Logo	PASS	Pass ✓	TN
1	waitUntil — Sign-up screen [VLM=false]	SKIPPED	Pass (skip)	Correct skip
2	click — Close Banner	PASS	Pass ✓	TN
3	click — Address Bar	PASS	Pass ✓	TN
4	click — Orta Sabanci Univ. A4 address	PASS	Pass ✓	TN
5	type — "little ceasers pizza"	PASS	Pass ✓	TN
6	click — Little Caesars Pizza card	PASS	Pass ✓	TN
7	click — Mixed (Medium) Pizza	FAIL (content_mismatch)	Fail X	TP ✓
8	click — Normal Dough	PASS	Pass ✓	TN
9	click — Less Pizza Sauce	PASS	Pass ✓	TN
10	click — Remove mushrooms & black olives	PASS	Pass ✓	TN
11	click — Add to cart	PASS	Pass ✓	TN
12	click — View your cart	PASS	Pass ✓	TN
13	click — Confirm basket	PASS	Pass ✓	TN
14	abstractVerify — Mixed (Medium) in checkout	FAIL (content_mismatch)	Fail X	TP ✓

Result: 2 failures detected, 2 real failures → Perfect precision and recall.

True Positive — Step 7: The click on "Mixed (Medium) Pizza" opened the "Mediterranean (Central)" pizza detail page. The visual observer correctly identified the wrong item in POST. The system operated in **vision-only mode** for the entire app (all 15 log buckets were empty per the nested format warning), demonstrating that the VLM pipeline alone is sufficient when the visual signal is unambiguous.

True Positive — Step 14: The abstract verification confirmed the wrong pizza persisted to checkout. The system reported Steps 7 and 14 as two **isolated failures** rather than a causal chain — consistent with the v2 finding that `find_likely_cause()` scores below 0.3 due to the 6-step temporal gap and action type difference (click vs. abstractVerification). This is a sub-optimality in chain reporting but does not affect detection accuracy.

App 7 — O Bilet · 9 steps · Log: nested (preassigned)

Scenario: Splash screen → category dashboard → set Bus + route Istanbul→Ankara (10 Apr) → search → open filter → apply company filter → select 22:00 AKSU trip → select Seat 12 + confirm → waitUntil Passenger Details form.

Step	Action / Element	System Result	Ground Truth	Assessment
0	waitUntil — App Logo	PASS	Pass ✓	TN
1	waitUntil — Category Dashboard	PASS	Pass ✓	TN
2	click — Bus tab + set parameters	PASS	Pass ✓	TN
3	click — SEARCH button	PASS	Pass ✓	TN
4	click — FILTER button	PASS	Pass ✓	TN
5	click — Apply multi-company filter	PASS	Pass ✓	TN
6	click — 22:00 AKSU trip	PASS	Pass ✓	TN
7	click — Seat 12 + Confirm and Continue	FAIL (system_error)	Fail ✗	TP ✓
8	waitUntil — Passenger Details Input	FAIL (element_missing)	Fail ✗	TP ✓

Result: 2 failures detected, 2 real failures → Perfect precision and recall.

True Positive — Step 7 (Root Failure): The system correctly identified the 504 Gateway Timeout on `POST /api/v3/bus/reserve-seat`. Log observer flagged `[NET_FAIL, SYS_ERR]` and the decider classified `system_error`. Visual: loading overlay persisted in POST (reservation did not complete).

True Positive — Step 8 (Cascading Failure): The waitUntil for Passenger Details form correctly failed. Log: `UI_STALL: Navigation to PassengerDetailsActivity blocked by failed reservation dependency`. The causal chain `Step 7 → Step 8` was correctly constructed. This is the cleanest two-step chain in the dataset.

4. Part B — Group 411 Traces: Case-by-Case Breakdown

Case 1 — Passo App · 11 steps · No unified log (step-level analysis)

Scenario: Verify Futbol category → click Futbol → tap Search → type "Eskişehirspor" → click match → click Bilet AI → increase ticket count → click Bilet AI (block selection) → select block 102 → click Koltuk Seç → click occupied (black) seat.

Ground truth per 411 reference output: Only Step 10 (their step_index 11: occupied seat click → no state change) is a real failure.

Step	Action / Element	System Result	Ground Truth	Assessment
0	abstractVerify — Futbol category visible	PASS	Pass ✓	TN
1	click — Futbol category	PASS	Pass ✓	TN
2	click — Search button	PASS	Pass ✓	TN
3	type — "Eskişehirspor"	PASS	Pass ✓	TN
4	click — Eskişehirspor - Tire 2021 FK	PASS	Pass ✓	TN
5	click — Bilet Al button	FAIL (action_failed)	Pass ✓	FP X
6	click — (+) ticket quantity	PASS	Pass ✓	TN
7	click — Bilet Al → block selection	PASS	Pass ✓	TN
8	click — Block 102	PASS	Pass ✓	TN
9	click — Koltuk Seç button	PASS	Pass ✓	TN
10	click — Occupied (black) seat	FAIL (action_failed)	Fail X	TP ✓

Result: 2 failures detected, 1 real failure → 1 TP + 1 FP.

True Positive — Step 10: The click on the black (occupied) seat produced no state change between PREV and POST. The system correctly classified this as **action_failed**. The 411 reference confirms: "No visual change detected after clicking the seat. (Root Cause: The click action may not have been registered, or the seat is not selectable.)" Both systems agree on this failure.

False Positive — Step 5: Our system observed the **Bilet Al** button becoming "inactive (grayed out)" with no navigation or content change in POST. However, the 411 reference output (step_index 6) shows a PASS: "The 'Bilet Al' button was clicked, leading to a change in the UI where ticket categories are now displayed." The discrepancy indicates a **VLM observation error** — our vision observer misread the POST screenshot state, interpreting the ticket category list appearance as a button grayout rather than a UI expansion. This is a false positive caused by visual hallucination in the observer, not a decider rule error.

Implication: The observer's **semantic_screen_summary** for Step 5 was insufficient to override the element-level misreading. This suggests the observer prompt's SELECTED STATE and navigation detection clauses need strengthening for collapsible/expandable UI elements that replace the trigger button with new content.

Case 2 — Bank App · 8 steps · Log: inline (g411 format)

Scenario: Verify balance (5250 TL) → navigate to Para Gönder → enter recipient "Tuna" → enter amount "300" → click Transfer → see Transfer Successful screen → navigate to dashboard → abstractVerify balance.

Ground truth per user annotation: Steps 4, 6, and 7 are real failures. Step 6 is a "systematic problem not about rules."

Step	Action / Element	System Result	Ground Truth	Assessment
0	abstractVerify — balance 5250.00 TL	PASS	Pass ✓	TN

Step	Action / Element	System Result	Ground Truth	Assessment
1	click — Para Gönder	PASS	Pass ✓	TN
2	type — "Tuna" in Alıcı adı	PASS	Pass ✓	TN
3	type — "300" in amount field	PASS	Pass ✓	TN
4	click — Transfer button	FAIL (system_error)	Fail ✗	TP ✓
5	see — Transfer Successful screen	PASS	Pass ✓	TN
6	navigate — Back to dashboard	PASS	Fail ✗	FN ✗
7	abstractVerify — balance unchanged	FAIL (element_missing)	Fail ✗	TP ✓

Result: 2 failures detected, 3 real failures → 2 TP + 0 FP + 1 FN.

True Positive — Step 4: The Transfer button triggered a 500 Internal Server Error. Log observer flagged [NET_FAIL, SYS_ERR]. The system correctly identified this as system_error despite the visual POST screen showing a "Transfer Successful" confirmation — the log evidence overrode the misleading visual (log can only override PASS→FAIL, never FAIL→PASS — functioning as intended).

True Positive — Step 7: The abstract verification element Bakiyeyi doğrula was not visible in either PREV or POST screens. The system correctly classified this as element_missing. This reflects the downstream consequence of Step 6's missed failure — the verification could not proceed because the expected state was never established.

False Negative — Step 6 (Systematic Problem):

Case 3 — Ticket App · 11 steps · Log: inline (g411 format)

Scenario: Verify Login screen → login (enter credentials) → click Theater category → click Sports category → click Details (Madrid Falcons) → click Select Seats → click occupied seat A3 → click available seat A1 → click Go to Cart → verify seat + price → click Confirm & Pay.

Ground truth per 411 reference output: Steps 2, 3, 6, and 10 are real failures.

Step	Action / Element	System Result	Ground Truth	Assessment
0	abstractVerify — Login screen present	PASS	Pass ✓	TN
1	click — Enter credentials + Login	PASS	Pass ✓	TN
2	click — Theater category	FAIL (action_failed)	Fail ✗	TP ✓
3	click — Sports category	FAIL (system_error, caused by Step 2)	Fail ✗	TP ✓
4	click — Details (Madrid Falcons)	PASS	Pass ✓	TN

Step	Action / Element	System Result	Ground Truth	Assessment
5	click — Select Seats	PASS	Pass ✓	TN
6	click — Occupied seat A3	FAIL (action_failed)	Fail X	TP ✓
7	click — Available seat A1	PASS	Pass ✓	TN
8	click — Go to Cart	PASS	Pass ✓	TN
9	abstractVerify — seat A1, price \$99.50	PASS	Pass ✓	TN
10	click — Confirm & Pay	FAIL (system_error, caused by Step 6)	Fail X	TP ✓

Result: 4 failures detected, 4 real failures → Perfect precision and recall.

True Positive — Step 2: Click on Theater category produced no visual change between PREV and POST. Log confirmed: "User tapped Theater." — the action was registered but the UI did not navigate. Classified as **action_failed**. Matches 411 reference.

True Positive — Step 3 (Cascading): Sports category navigation showed a RenderFlex overflow (869 pixels on the right side). Log: console rendering exception. Classified as **system_error**. The causal chain **Step 2 → Step 3** was correctly constructed — the cascade is plausible because the non-navigation at Step 2 may have left the app in an unstable UI state that surfaced on the next navigation.

True Positive — Step 6: Click on occupied seat A3 returned **ERROR: Seat unavailable**. No UI state change. Classified as **action_failed**. Matches 411 reference exactly.

True Positive — Step 10 (Cascading): **POST /api/v1/payment → 500 Internal Server Error**. Classified as **system_error**. The causal chain **Step 6 → Step 10** was constructed — the occupied-seat error likely corrupted the cart/reservation state, preventing payment completion.

Two failure chains correctly identified:

- **Chain 1:** **Step 2 → Step 3** (UI non-response → rendering error)
- **Chain 2:** **Step 6 → Step 10** (seat error → payment server error)

This is the best-performing 411 case: 100% precision and recall with correctly structured chains. The inline log format (**[g411]**) provided good evidence quality, and the 411 reference output confirms all four failures align with our detections.

Case 4 — Rent-a-Car App · 11 steps · Log: format=none (no unified log; inline logs for some steps)

Scenario: Verify main screen → select cities (Ankara→Izmir) → select dates (14–17 April) → search vehicles → verify vehicle list → click Kirala for Mercedes-Benz G-Class → navigate to payment screen → verify reservation details → enter promo code "DISCOUNT20" → click Uygula (apply) → click Onayla ve Öde (pay).

Ground truth per user annotation: Steps 5, 8, 9, and 10 are real failures. Step 8 fails because the discount was claimed applied by the UI but the total price did not update — a visual inconsistency the observer should have caught.

Step	Action / Element	System Result	Ground Truth	Assessment
0	abstractVerify — main screen loaded	PASS	Pass ✓	TN
1	click — Select cities Ankara/Izmir	PASS	Pass ✓	TN
2	click — Select dates 14/17 April	PASS	Pass ✓	TN
3	click — Search vehicles	PASS	Pass ✓	TN
4	abstractVerify — vehicle list displayed	PASS	Pass ✓	TN
5	click — Kirala (Mercedes-Benz G-Class)	FAIL (action_failed)	Fail ✗	TP ✓
6	navigate — to payment screen	PASS	Pass ✓	TN
7	abstractVerify — total ₺17,000 for 3 days	PASS	Pass ✓	TN
8	type — Promo code "DISCOUNT20"	PASS	Fail ✗	FN ✗
9	click — Uygula (apply discount)	FAIL (content_mismatch)	Fail ✗	TP ✓
10	click — Onayla ve Öde (pay)	FAIL (system_error, caused by Step 9)	Fail ✗	TP ✓

Result: 3 failures detected, 4 real failures → 3 TP + 0 FP + 1 FN.

True Positive — Step 5: The Kirala action for Mercedes-Benz G-Class was blocked by a backend requirement: "special corporate account required for intercity drop-off in Izmir." Log observer flagged [SYS_ERR]. Classified as **action_failed** — visually the button appeared active but the action did not proceed.

True Positive — Step 9: The discount application showed the success message "Tebrikler! %20 indirim uygulandı!" but the total remained ₺17,000.00 unchanged. Classified as **content_mismatch** — the expected new total (₺13,600) was not confirmed in POST. Correctly linked as caused by Step 5 (the intercity restriction created an inconsistent reservation state).

True Positive — Step 10: POST /payment → 504 Gateway Timeout on Izmir regional gateway. Classified as **system_error**. Correctly chained from Step 9.

False Negative — Step 8: Entering DISCOUNT20 into the promo code field appeared visually successful — the text was correctly entered into the input. Our system reported PASS. The ground truth requires FAIL because the discount was claimed applied by the UI (the success message "Tebrikler! %20 indirim uygulandı!" was visible) yet the total price of ₺17,000 remained unchanged in the same view. This price-discount

inconsistency was present in the POST screenshot and should have been detectable by the Vision Observer: a 20% discount on ₺17,000 yields an expected total of ₺13,600, but ₺17,000 was still displayed. The observer reported the promo code entry as successful without cross-checking whether the displayed total reflected the claimed discount. This is a **visual observation failure** — the evidence was on-screen but not caught.

Case 5 — Wikipedia Website · 5 steps · No log (format=none)

Scenario: Verify Vikipedi page present → click top-left menu → click Hakkımızda → click "138 dil" language option → click "Osmanlı padişahı 3. Murad tahta çıktı seç."

Ground truth per 411 reference output: Step 4 is a real failure (instruction target not present in any visible UI element).

Step	Action / Element	System Result	Ground Truth	Assessment
0	abstractVerify — "Vikipedi" visible	PASS	Pass ✓	TN
1	click — Top-left menu icon	PASS	Pass ✓	TN
2	click — Hakkımızda	PASS	Pass ✓	TN
3	click — "138 dil" option	PASS	Pass ✓	TN
4	click — Osmanlı padişahı 3. Murad...	FAIL (element_missing)	Fail ✗	TP ✓

Result: 1 failure detected, 1 real failure → Perfect precision and recall.

True Positive — Step 4: The instruction target "Osmanlı padişahı 3. Murad tahta çıktı seç." is lexically isolated from the current page context (Wikipedia Hakkımızda page showing general information). The element was absent from both PREV and POST screenshots. Classified as `element_missing`. The 411 reference classifies this as an "Instruction Error" with root cause: "The instruction seems to be unrelated to the current Wikipedia page context, possibly due to a scenario mismatch or incorrect step." Both systems agree: the target element is absent and the step fails. Operating in vision-only mode (no log data) was sufficient for this unambiguous absence detection.

5. Precision, Recall & F1 by App

5A. Our Own Traces

App	Analyzed Steps	GT Failures	Predicted	TP	FP	FN	TN	Precision	Recall	F1
Migros	8	1	1	1	0	0	7	100%	100%	100%
Bitaksi	8	0	0	0	0	0	8	N/A	N/A	N/A
Clock	7	0	0	0	0	0	7	N/A	N/A	N/A
Flight App	22	6	6	6	0	0	16	100%	100%	100%
QNB	13	1	1	1	0	0	12	100%	100%	100%

App	Analyzed Steps	GT Failures	Predicted	TP	FP	FN	TN	Precision	Recall	F1
Yemeksepeti	14	2	2	2	0	0	12	100%	100%	100%
O Bilet	9	2	2	2	0	0	7	100%	100%	100%
TOTAL	81	12	12	12	0	0	69	100.0%	100.0%	100.0%

For apps with 0 ground truth failures, precision is undefined (no positive examples). Bitaksi and Clock contribute 0 TP and 0 FP — pure TN performance.

5B. Group 411 Traces

App	Analyzed Steps	GT Failures	Predicted	TP	FP	FN	TN	Precision	Recall	F1
Passo	11	1	2	1	1	0	9	50.0%	100.0%	66.7%
Bank App	8	3	2	2	0	1	5	100.0%	66.7%	80.0%
Ticket App	11	4	4	4	0	0	7	100.0%	100.0%	100.0%
Rent-a-Car	11	4	3	3	0	1	7	100.0%	75.0%	85.7%
Wikipedia	5	1	1	1	0	0	4	100.0%	100.0%	100.0%
TOTAL	46	13	12	11	1	2	32	91.7%	84.6%	88.0%

5C. Combined Summary

Dataset	Analyzed	GT Fail	Predicted	TP	FP	FN	TN	Precision	Recall	F1
Our Traces	81	12	12	12	0	0	69	100.0%	100.0%	100.0%
Group 411	46	13	12	11	1	2	32	91.7%	84.6%	88.0%
COMBINED	127	25	24	23	1	2	101	95.8%	92.0%	93.9%

6. Error Classification

6.1 False Positive (1 total)

FP #	App	Step	Trigger	Root Cause Category
1	Passo App	Step 5	Vision observer reported Bilet Al button became "inactive (grayed out)" with no navigation; 411 reference shows ticket categories appeared (PASS)	VLM observation error — observer misread UI state (ticket category expansion misidentified as button deactivation)

Analysis: The false positive originates in the Vision Observer, not the Logic Decider. The observer's element-level description of POST failed to identify the ticket category list that appeared after clicking `Bilet A1`. This type of error — confusing a button that triggers content expansion with a button that becomes disabled — indicates the observer prompt's `SELECTED STATE` and navigation detection clauses need explicit handling for cases where the trigger element is replaced by revealed content rather than remaining visible. The `semantic_screen_summary` (1-2 sentence summary) should have caught the navigation-to-new-content, but did not provide sufficient signal in this case.

6.2 False Negatives (2 total)

FN #	App	Step	System Result	Ground Truth	Root Cause Category
1	Bank App	Step 6	PASS (dashboard loaded)	FAIL (balance unchanged after "successful" transfer)	Cross-step state inconsistency — requires tracking expected state across non-adjacent steps (Step 0 balance, Step 3 amount, Step 5 success claim, Step 6 balance comparison)
2	Rent-a-Car	Step 8	PASS (promo code entered)	FAIL (discount claimed applied but total price unchanged)	Visual observation failure — discount success message and unchanged ₺17,000 total were both visible in POST; observer reported text entry success without verifying the price reflected the claimed discount

FN Pattern Analysis: The two false negatives arise from different root causes:

- **Bank App Step 6** — the failure produces no single-step visual evidence; it is only detectable by comparing the balance at Step 6 against the expected post-transfer value derived from Steps 0 and 3. This is a genuine **cross-step state inconsistency** that requires expected-state propagation across non-adjacent steps.
- **Rent-a-Car Step 8** — the failure was visually detectable within the step itself: the UI simultaneously displayed a discount-applied success message and an unchanged total price of ₺17,000. This is a **visual observation failure** — the Vision Observer focused on confirming the promo code was entered rather than verifying that the displayed total was consistent with the claimed discount.

These two cases are structurally different: Bank App Step 6 is an architectural gap (cross-step tracking not yet implemented); Rent-a-Car Step 8 is an observer prompt gap (price-discount consistency check missing from the observer's evaluation criteria).

6.3 Sub-optimality (Not Detection Errors)

Issue	App	Description
Missed causal chain	Yemeksepeti	Steps 7 and 14 both detected but reported as ISOLATED rather than a chain (Step 7 → Step 14). <code>find_likely_cause()</code> scored below 0.3 due to 6-step temporal gap and click vs. <code>abstractVerification</code> type mismatch. Both steps were correctly detected.

Issue	App	Description
Causal link to FN	Rent-a-Car	The chain Step 5 → Step 9 → Step 10 is partially correct. The actual root chain should be Step 8 → Step 9 → Step 10 (missed price-discount inconsistency → amount mismatch → payment timeout). The system constructed a valid but non-root chain that traces to the earlier Step 5 blockage.
Bank FN cascade	Bank App	Step 7 (element_missing) is downstream of Step 6 (FN). Because Step 6 was missed, Step 7 appears as an isolated failure rather than the second step in a chain Step 6 → Step 7 . The chain structure is recoverable if Step 6 detection is added.

7. Key Takeaways

7.1 Improvements from v32.x to v33.0 (Our Traces)

Issue (v2 report)	v2 Count	v33 Count	Status
Log-step index mismatch FPs (Clock, QNB)	4 FPs	0	Fully resolved
Rule 2 over-application FPs (Migros)	2 FPs	0	Fully resolved
Invalid failure chains (Migros, QNB)	2 spurious chains	0	Fully resolved
Missed causal chain (Yemeksepeti 7→14)	1	1	Still present (sub-optimality)
Total FPs	6	0	−100%
Total FNs	0	0	No change

7.2 Performance Dimensions Summary

Dimension	Our Traces	Group 411	Combined	Verdict
Recall / failure detection	100%	84.6%	92.0%	Excellent on owned data; strong on external
Precision / noise	100%	91.7%	95.8%	Excellent; single VLM obs. error on external
F1 Score	100%	88.0%	93.9%	High across both datasets
Log-integrated detection	100% (with logs)	91.7%	—	Log evidence consistently improves recall
Vision-only detection	100% (Yemeksepeti)	100% (Wikipedia)	—	VLM sufficient when visual signal is strong

Dimension	Our Traces	Group 411	Combined	Verdict
Causal chain accuracy	Mixed (1 missed chain)	Good (2/3 chains correct)	—	Correct linking in most cases
Cross-step consistency	Not applicable	0% (2 FNs)	—	Current architectural gap

7.3 Generalization Assessment

The system was developed and tuned on the proprietary traces (Migros, Bitaksi, Clock, Flight App, QNB, Yemeksepeti, O Bilet). Applying it to the independently designed Group 411 test cases — without any adaptation — produced an F1 of **88.0%**, a **12-point drop from perfect score**. This gap is explained entirely by two limitations — a cross-step state tracking gap (Bank App Step 6) and a visual observation failure where a price-discount inconsistency was on-screen but not caught (Rent-a-Car Step 8) — not by prompt tuning or model behavior. The 1 false positive from the Passo VLM observation error is a random noise event, not a systematic flaw.

The 411 test cases were intentionally designed with subtle, realistic faults ("a sequence of successful, normal steps that ultimately leads to one intentional, critical bug"). The system detected 11 of 13 such intentional bugs — a strong result for a first-pass evaluation on unseen data.

8. Recommendations

Rec 1 — Multi-step state propagation for abstractVerification steps (addresses both FNs)

Both false negatives (Bank App Step 6, Rent-a-Car Step 8) require comparing a value from a previous step against the current step's observations. Implement a **state scratchpad** in **GraphState** that records key numerical values extracted by the Vision Observer from prior steps (balances, amounts, totals). When the Logic Decider encounters a step where the **action** is **abstractVerification** or the instruction references a value from a prior step, inject the relevant scratchpad entries as **[STATE CONTEXT]** alongside the regular evidence. This would enable cross-step inconsistency detection without restructuring the pipeline.

Estimated impact: Eliminates FN at Bank App Step 6 (balance comparison). May also close Rent-a-Car Step 8 if the expected post-entry validation state is tracked.

Rec 2 — Observer prompt: explicit handling of content-expansion patterns (addresses Passo FP)

Add a clause to the Vision Observer prompt covering **trigger-reveal patterns**: when a button is clicked and new content appears in POST that replaces or overlays the trigger area, the observer must report the revealed content as the primary POST state — not infer the trigger button's final state from visual ambiguity. Example guidance: "If clicking a button causes a new panel, list, or overlay to appear, describe what appeared — do not assess whether the button is active or inactive unless it is still clearly visible in POST." This directly targets the Bilet AI misread.

Estimated impact: Eliminates the Passo FP. Prevents similar misreads on accordion menus, expandable forms, and ticket category selectors.

Rec 3 — Strengthen causal chain scoring for abstractVerification steps (addresses Yemeksepeti sub-optimality)

abstractVerification steps that fail should be treated as likely downstream of any earlier **content_mismatch** failure involving the same logical entity. Add a keyword-similarity check between the failed **abstractVerification** instruction and prior failure **root_cause** strings. A cosine similarity above 0.5 should override the 0.3 threshold in **find_likely_cause()** when the types are **click** → **abstractVerification**. This was identified in the v2 report and remains relevant.

Rec 4 — Observer prompt: verify price-discount consistency for promo code steps (addresses Rent-a-Car FN)

When a step involves entering a promo code, coupon, or discount value, the Vision Observer should explicitly check whether the displayed total price is consistent with the claimed discount. Add a clause to the observer prompt: "If the POST screen shows a discount success message alongside a price or total value, verify that the displayed total reflects the stated discount percentage applied to the pre-discount amount. If the total is unchanged despite the success message, report this as a price-discount inconsistency." This targets the Step 8 miss directly — both the discount message and the unchanged ₺17,000 were on-screen but the observer did not cross-check them.

Rec 5 — Confidence distribution review for 411 traces

All 131 analyzed steps in this evaluation were classified as **High confidence (≥0.85)** with average confidence of 0.92–0.95 across apps. Given that 3 steps were incorrectly evaluated (1 FP, 2 FN), the confidence scores do not distinguish between correct and incorrect predictions in this dataset. Consider adding a calibration pass that penalizes confidence when: (a) log data is absent (**has_logs=False**) for an action type that typically generates backend events (type, click-on-submit, abstractVerify), or (b) the visual observer's **semantic_screen_summary** contradicts its **post_screen_changes** at the element level. Calibrated confidence would allow downstream consumers to identify the 3 uncertain steps rather than treating all high-confidence outputs uniformly.

9. Dataset Coverage Summary

App / Case	Source	Log Format	Mode	Steps	GT Fail	TP	FP	FN
Migros	Our traces	nested	preassigned	8	1	1	0	0
Bitaksi	Our traces	flat_assertions	preassigned	8 (1 skip)	0	0	0	0
Clock	Our traces	flat_indexed	preassigned	7	0	0	0	0
Flight App	Our traces	raw	dynamic/keyword	22 (3 skip)	6	6	0	0

App / Case	Source	Log Format	Mode	Steps	GT Fail	TP	FP	FN
QNB	Our traces	flat_indexed	preassigned	13	1	1	0	0
Yemeksepeti	Our traces	nested (empty)	preassigned	14 (1 skip)	2	2	0	0
O Bilet	Our traces	nested	preassigned	9	2	2	0	0
Passo App	Group 411	inline g411 / none	per-step	11	1	1	1	0
Bank App	Group 411	inline g411 / none	per-step	8	3	2	0	1
Ticket App	Group 411	inline g411	per-step	11	4	4	0	0
Rent-a-Car	Group 411	inline g411 / none	per-step	11	4	3	0	1
Wikipedia	Group 411	none	none	5	1	1	0	0
TOTAL				127	25	23	1	2

Report generated for Describer-Decider v33.0. Evaluation date: 2026-04-28. Our traces: 7 proprietary apps across Turkish e-commerce, transport, finance, and food delivery domains. Group 411 traces: 5 independently designed test cases (Passo ticketing, bank transfer, event ticketing, rent-a-car, Wikipedia web)